PROJECT #3 – VISUALIZE

Client: Federal Highway Administration (FHWA)

Client Prompt:

Transportation systems are among the most critical components of the economy and daily life. To support informed decision-making, the United States invests heavily in collecting extensive data to understand transportation usage and system performance. Two highly influential datasets in this field are the National Household Travel Survey (NHTS) and the Next Generation Simulation (NGSIM) Data.

These two datasets have been made available to your company. NHTS provides insights into transportation usage characteristics, while NGSIM focuses on driving behavior. Because you are the Engineering Intern with the most programming experience in Python, your PE has asked you to help organize and visualize the data prior to a meeting of the transportation planning group in your firm. Your team has been provided with two different CSV files and two Metadata files that explain the content of the data sets.

Specifically, your PE asks you to complete the following tasks:

1. Create clean and simple visualizations of the important features and trends in the data
    a. Specifically, your PE has asked that the following visualizations are present:
        i. A bar chart comparing categories within a specific variable
        ii. A histogram showing the distribution of a specific variable
        iii. A boxplot showing the distribution of a variable in each category
        iv. At least two different time-series plots showing and analyzing driving behavior
    b. Your PE has not specified which variables to plot, only that the variables plotted need to be important and representative to your transportation planning group. You are allowed to select any variables so long as they provide insight into transportation trends and driving behavior in the US. You must use different variables for different plots. The metadata files (codebooks) given to you provide additional information about what is in NHTS and NGSIM datasets.
    c. At least one of the plots should demonstrate that you can alter the base level plots available in Python. You can change text font, plot point or line colors, etc. At least one of the plots must show evidence that you have used the matplotlib or seaborn library to customize your plot.
    d. All plots must be properly titled, with clearly labelled axes and legends where necessary.
2. Your client also asks you to provide a Simulation study using the Intelligent Driver Model (IDM), to visualize and simulate vehicle behavior and vehicle trajectories.
    a. You need to select a specific trajectory and simulate the follower based on the NGSIM data for the leader in the trajectory pair.
    b. Simulate acceleration of the follower and create a plot showing the acceleration of the leader, the follower (from NGSIM data) and the follower simulated data using the IDM model
3. Document the amount of time you and your other Engineering Intern friends spend working on each task. Your PE is interested in documenting how long it takes to perform the data analysis in Python so they can better value your time and compare it to how long the analysis takes in

Excel. Your PE is interested in investing in Python training for other EIs in your firm and wants to see how much time Python might save them in the future.

The summary of the deliverables requested by your PE and the deadline are listed below.

Deliverables:

1. GitHub
    a. GitHub Repository: Provide a link to a repo containing your code, data, and a README. The README must summarize the project and provide a "User Guide" for the sequential input prompts.
2. Data files used in the Python Code file
3. Python code file that documents your code used to generate your solution
4. A well-written scope of work related to the client prompt that defines the tasks undertaken by your group to complete the project. Scope of work must contain the following items:
    a. An identification of the variables you will be examining in your graphs using the provided code book information on the NHTS and NGSIM.
    b. An accurate explanation of the NHTS or NGSIM variables you plan on including representation in the data set.
5. A Gantt chart of the activities completed as part of the project
6. A timesheet detailing hours spent on each activity and by whom within the group.
    a. Details provided here: Engineering Timesheet.docx
7. An annotated code document that explains each line of code in detail
8. A written document that summarizes the findings contained in the code that can be handed to your PE to be used at a later date.
    a. Your report must include the following elements:
        i. Introduction: Background on the FHWA project and your objectives
        ii. Methods:
            1. A description of the variables you selected and why these were selected by your team
            2. A description of how each plot was constructed
            3. A description of any method you have used
        iii. Results and Discussion:
            1. A discussion of the key findings in each of your plots
            2. A discussion of how each of these key findings could be relevant to the transportation industry
        iv. References: proper intext citations and a reference list

All files must be submitted to Canvas by Tuesday March 31st , 2026 at 5:00pm CT.

Skills Students will learn:

- How to summarize data/code in written reports for both technical and non-technical audiences
- How to distinguish between several types of graphs
- How to choose appropriate graphs for the analysis needed

- How to create graphs using Python and the matplotlib library
- How to do simulation studies

Rubric:

Component	Description	Points Possible	Points earned	Instructor Feedback
Jupyter notebook containing Python code (.ipynb)	A Jupyter Notebook that documents the code used to automate the "Mix Design" tab. It must include sequential user prompts, calculations and must be tested for four mix designs.	35		
Github Link	A link to a repository containing the code, data, and a READMe file that summarizes the project and identifies each file	10		
GANTT chart	See GANTT rubric for more details	10		
Annotated Code Document	Please see the rubric for the annotated code document on Canvas	10		
Engineering Timesheet	Timesheet is well organized and contains all the required components. Timesheet includes relevant tasks, a clear distribution of work and describes each task completed.	10		
Scope of Work	See scope of work rubric for more details	30		
Technical report	Report must include the following elements: Introduction: Background on the FHWA project and your objectives Methods: A description of the variables you selected and why these were selected by your team A description of how each plot was constructed A description of any method you have used Results and Discussion: A discussion of the key findings in each of your plots A discussion of how each of these key findings could be relevant to the transportation industry References: proper intext citations and a reference list	55		
TOTAL		160		